

To test the scanners, they created life-size 3D hand models, complete with all five fingerprints, using a high-resolution 3D printer that can produce the same ridges and valleys as a real finger's.

Technology to make 2D, 3D printing super fast

Scientists have found a way to increase the speed of 2D and 3D printing by up to 1000 times, which could allow surgeons to non-invasively obtain images of organs in real time. Using a space-charge-controlled KTN beam deflector—a kind of crystal made of potassium tantalate and potassium niobate—with a large electro-optic effect, researchers have found that scanning at a much higher speed is possible.

According to Shizhuo Yin, professor at School of Electrical Engineering and Computer Science, Pennsylvania State University, USA, when crystal materials are applied to an electric field, these generate uniform reflecting distributions that can deflect an incoming light beam.

To overcome this issue, Yin and his team eliminated the electric field-induced phase transition in a nano-disordered KTN crystal by making it work at a higher temperature. They not only went beyond Curie temperature, they went beyond the critical end point in which a liquid and its vapour can coexist. This increased the scanning speed from the microsecond range to the nanosecond regime and improved high-speed imaging, broadband optical communications and ultra-fast laser display and printing.

Air-purifier that is sucking up China's smog and turning it into jewellery

Dutch artist Daan Roosegaarde and his team have erected a seven-metre-tall air-purifier in Beijing, China. The so-called Smog Free Tower purportedly treats 30,000 cubic metres of air per hour, collecting more than 75 per cent of two kinds of pollutants, PM2.5 and PM10, that contribute to smog.

The Smog Free Tower not only sucks up pollution, it also turns particulate matter into jewellery by compressing 1000 cubic metres of polluted air into rings, which people can purchase.



Compressed smog particles are sealed in a resin cube after being compressed for 30 minutes; resulting objects can be used to make jewellery such as rings and cufflinks (Image courtesy: shanghaiist.com)

The massive structure has its roots in a Kickstarter campaign that raised €113,153 in August 2015. It appeared first in Rotterdam, The Netherlands, last year before making its way east. With support from China's Ministry of Environmental

Protection, the structure will travel to three additional cities in the country after its stint in Beijing.

Solar panels that can generate clean drinking water

An Arizona, USA, startup called Zero Mass Water is manufacturing solar-powered personal water production systems that collect water vapour from heated air. The water vapour then becomes potable water after it has had a chance to cool down. The technology branded Source pulls water out of thin air to cover the daily needs of a family of four.



Source can help fulfil daily water needs of families in developing countries (Image courtesy: www.inhabitat.com)

Each Source unit looks like a solar panel resting on top of a metal box. Source panels do not need an external power source to run, like other solar panels that also purify water. The device essentially uses sunlight to produce electricity and heat, which allows a set of proprietary materials to passively catch humidity in the air.

Jaguar unveils concept electric car

Tata Motors-owned luxury car brand Jaguar has made its way into the electric car arena with the launch of its new concept car, I-Pace, simultaneously in London, UK, and Los Angeles, USA. The car can cover a distance of more than 500 kilometres on a single full charge.



Jaguar's designers capitalised on the versatility of compact electric power trains, using the space normally used by an engine and transmission to push the seats forward and the roof down (Image courtesy: www.jaguar.in)

Ian Callum, director of design, Jaguar Land Rover, said during the launch that was beamed live from Los Angeles through virtual reality to London, "The I-Pace Concept represents the next generation of electric vehicle design. It is a dramatic, future-facing cab-forward design with a beautiful interior—the product of authentic Jaguar DNA, electric technology and contemporary craftsmanship."

Scheduled for production in 2017, the all-electric battery-powered Jaguar SUV is expected to hit the roads towards the end of 2018.